

# Road Accident Simulation Report

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Accident Type: Vehicle-pedestrian collision  
Location: Bulhao, Labo, Camarines Norte  
Time of Accident: 1.74 seconds into simulation  
Cause: Reckless Driving  
Weather: Foggy  
Road Geometry: straight (Slope: 0.0°)

### Vehicles Involved:

- Car 1: Mass = 1206 kg, Initial Speed = 74.0 km/h
- Pedestrian: Mass = 70 kg

### Analysis:

This simulation demonstrates how speed, road conditions, and lighting contribute to accidents. Severity is classified based on impact force: minor (<50kN), moderate (50-150kN), severe (>150kN).

### Recommendations:

- Maintain current lighting standards
- Install anti-skid road surfaces
- Increase traffic police presence
- Install pedestrian crossings
- Implement point-based license system

### Systems Interaction Summary:

- Reckless driving combines aggressive human behavior with vehicle capabilities and environmental factors in systems theory, leading to high-risk interactions.

### Intervention Scenario Comparison:

- Baseline Severity Distribution: severe: 14, none: 4, moderate: 2
- Intervention Severity Distribution: severe: 9, moderate: 6, none: 5
- Baseline Avg Impact Force: 938230.12 N
- Intervention Avg Impact Force: 720528.06 N
- Avg Risk Score Reduction: 10.6% (8.39 -> 7.50)
- Avg Impact Force Reduction: 23.2%
- Interventions Applied: Road Surface Treatment

### Machine Learning Prediction:

Predicted Severity: Unknown (0.0% confidence)